

128 players enter a chess tournament. Each round, every remaining player is paired up; the loser is eliminated, the winner advances. This repeats until one champion remains.

1. How many players remain after round 1? Round 2? Continue this pattern until you reach 1 player. (5 points)
2. How many *rounds* does the whole tournament take? Express it as $\log_2(?) = ?$. (5 points)
3. How many *games* are played in round 1? Round 2? Add up the total games across all rounds — Can we represent this as a function of n ? (10 points)

Answers:

1. 64, 32, 16, 8, 4, 2, 1
2. $\log_2(128) = 7$ rounds
3. Round 1: 64 games, round 2: 32, ... total = $64+32+16+8+4+2+1 = 127$ games
This can be represented as $2^0+2^1+\dots+2^{(7-1)}$
Geometric progression: 1st term = 2^0 , common ratio=2, Number of terms = n
Sum = $2^n - 1$
In our case $n = 7$ rounds, so total number of games in 7 rounds = $2^7 - 1 = 127$

For part 3, if they did get to the total 127 without the generic formula for geometric progression, still allocate them 5 points. If they get to the formulation as function of n , then allocate 10 points.